Summary in Graph

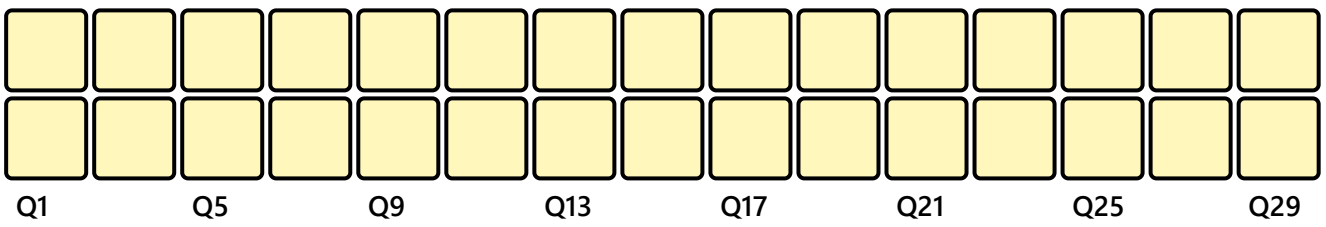
Exam Summary (GATE Overflow CS Test Series 2026 | Databases | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (6) out of 63 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following are correct? (Mark all the appropriate options)

A. A relation with only 2 attributes will always be in BCNF
B. A relation with only 2 attributes will always be in $3\mathrm{NF}$ but not necessarily in BCNF
C. A relation in $2\mathrm{NF}$ may have partial dependency involving prime attributes
D. For a relation in BCNF, if a functional dependency $A \rightarrow B$ exists, B must be a super key

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements regarding database normalization is/are correct? (Mark all the appropriate choices)

A. A relation is in Boyce-Codd Normal Form (BCNF) if and only if every determinant is a super key
B. A relation is in Fourth Normal Form (4NF) if and only if for every nontrivial multi-valued dependency $A \twoheadrightarrow B$, A is a super key of the relation
C. A multi-valued dependency (MVD) represents a dependency between attributes A, B and C in a relation such that for each value of A there is a set of values of B and a set of values for C and the set of values of B and C are independent of each other
D. A lossless-join dependency is a property of decomposition, which means that no spurious tuples are generated when relations are combined through a natural join operation

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are correct? (Mark all the appropriate options)

A. In a B^+ Tree, the structure of leaf nodes differs from the structure of internal nodes
B. In a B^+ Tree, some search field values from the leaf nodes are repeated in the internal nodes to guide the search
C. In a B^+ Tree, all non-root nodes are at least half full
D. In a B^+ Tree, every key value exist at the leaf level

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following is/are correct regarding a weak entity in Entity-Relationship model? (Mark all the appropriate options)

A. Every weak entity has to be related to a strong entity through a strong relationship
B. A weak entity cannot be uniquely identified by its own attributes
C. The participation of weak entity in its identifying relationship is total
D. A weak or non-identifying relationship exists between two entities when the primary key of one of the related entities contain a primary key component of the other related entities

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 09sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Let $A(a, b, c)$ and $B(d, e, f)$ be two relations in which d is the foreign key of B that refers to the primary key of A.

Which of the following four operations on A and B will not cause a violation of the referential integrity constraint? (Mark all the appropriate choices)

A. Insert into A
B. Delete from A
C. Insert into B
D. Delete from B

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are TRUE?
(Mark all the appropriate choices)

A. Index whose search key specifies the sequential order of the file is called a secondary index
B. Secondary index must be sparse
C. Primary index must be dense
D. In a clustered index, order of data records is the same as order of index entries

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Given the following relation R

R

A	B
1	NULL
1	NULL
NULL	1
4	NULL

The output of the following query will be _____

```
select sum(A) + avg(A) + sum(B) + avg(B) from R
```

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

From the following instance of a relation schema $R(A, B, C)$, we can conclude that:(Mark all the appropriate options)

A	B	C
1	4	5
5	6	4
1	3	5
5	2	2
2	4	3

A. A does not functionally determine B and B does not functionally determine C.
B. A does not functionally determine C and B does not functionally determine C.
C. C does not functionally determine B and B does not functionally determine A.
D. C functionally determines A and B does not functionally determine C.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are correct? (Mark all the appropriate choices)

A. For every relational algebra expression there is an equivalent expression in the relational calculus
B. For every safe tuple relational calculus expression there is an equivalent relational algebra expression
C. For every safe domain relational calculus expression there is an equivalent relational algebra expression
D. For every safe tuple relational calculus expression there is an equivalent safe domain relational calculus expression

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following relation R

R

A	B
1	NULL
1	NULL
NULL	NULL
3	NULL

The output of the following query will be ____

```
select count(*) + count(A) + count(B) from R
```

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Assume that relation R is defined over the attribute set A and relation S is defined over the attribute set B such that $B \subseteq A$. Let $C = A - B$, that is, C is the set of attributes of R that are not attributes of S. A division operator $R \div S$ defines a relation over the attributes C that consists of the set of tuples from R that match the combination of every tuple in S. Which of the following relational algebra query is equivalent to $R \div S$?

A. $\Pi_C(R) - \Pi_C(\Pi_C(R) \times S) - R)$
B. $\Pi_C(R) - \Pi_C(\Pi_C(R) \times S) - S)$
C. $\Pi_C(R) - \Pi_A(\Pi_C(R) \times S) - S)$
D. $\Pi_C(R) - \Pi_C(\Pi_B(S) \times R) - R)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the relations $R(A, B, C)$ and $S(B, C, D)$ and the following SQL queries.

I. Select distinct $*$ from R, S;
II. Select distinct $*$ from R, S where $R.C = S.C$;
III. Select distinct $*$ from R, S where $R.C = S.C$ and $R.B = S.B$;
IV. Select distinct $*$ from R, S where $R.C! = S.C$;

Match the following Table where the first column refers to the four given SQL queries

$i.$	$a.$	Equi Join
$ii.$	$b.$	Natural Join
$iii.$	$c.$	Theta Join
$iv.$	$d.$	Cross Product

A. $i-d, ii-a, iii-c, iv-b$
B. $i-b, ii-a, iii-d, iv-c$
C. $i-d, ii-a, iii-b, iv-c$
D. $i-c, ii-a, iii-b, iv-d$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following relation instance.

Student

Roll	FirstName	LastName	Weight
101	Rahul	Soni	40
102	Anam	Patel	55
103	Donald	Trump	75
104	Anuj	Sahu	45
105	Kim-jong	Un	105
106	Alex	Carry	50

Enrol

Roll	CourseId
101	1000
101	1001
101	1006
102	1000
103	1006
103	1001

Which of the following queries will correctly retrieve the course IDs of all the courses taken by students weighing more than 45 units?
(Mark all the appropriate options)

A. $\{t \cdot \text{CourseId} \mid \text{Enrol}(t) \wedge \exists s(\text{Student}(s) \wedge s.Roll = t.Roll \wedge s.weight > 45)\}$
B. $\Pi_{\text{CourseId}}(\sigma_{weight>45}(\text{Student} \bowtie_{Student.Roll=Enroll.Roll} Enroll))$
C. $\Pi_{\text{CourseId}}(\sigma_{weight>45 \wedge Student.Roll=Enroll.Roll}(\text{Student} \times Enroll))$
D. Select Enrol.CourseId from Enrol,Student where Student.weight > 45

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a $m:n$ relationship and an ER model using it. Which of the following is/are correct for converting this relationship to a relational instance? (Mark all the appropriate options)

A. If participation of both the entities are mandatory then we only need 2 relations
B. If participation of both the entities are mandatory then we need 3 relations
C. If participation of only one entity is mandatory then we only need 2 relations
D. If participation of both the entities are optional then we need 3 relations

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Given the following relation instance.

X	Y	Z
1	0	5
2	1	2
1	4	5
3	4	3
4	3	4
3	5	3

Which of the following functional dependencies are satisfied by the above relational instance $R(X, Y, Z)$? (Mark all the appropriate options)

A. $XY \rightarrow Z$ and $Z \rightarrow Y$
B. $YZ \rightarrow X$ and $Y \rightarrow Z$
C. $YZ \rightarrow X$ and $X \rightarrow Z$
D. $XZ \rightarrow Y$ and $Y \rightarrow X$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 2 | Penalty: 0 | Databases

The maximum number of super keys for the relation schema $R(A_1, A_2, A_3, A_4, A_5)$ with two candidate keys A_1 and A_2 is _____

Your Answer: | Correct Answer: 24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The maximum number of super keys for the relation schema $R(A_1, A_2, A_3, A_4, \ldots, A_n)$ with A_1A_2 and A_1A_3 as the candidate key is _____

A. $2^{n-1} + 2^{n-1} - 2^{n-2}$
B. $2^{n-1} - 2^{n-3}$
C. $2^{n-2} - 2^{n-2} - 2^{n-3}$
D. $2^{n-1} + 2^{n-2} - 2^{n-3}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following relations.

Emp

EmpId	DeptNo	EmpName	Salary
1	5	Alex	10,000
2	2	Steve	20,000
3	1	Bob	15,000
4	2	Smith	18,000
5	4	Rakul	24,000
6	3	Rhea	17,000

Dept

DeptId	DeptName
1	CSE
2	ECE
3	ECE
4	IT
5	ME

Here, Emp(DeptNo) is the foreign key referring to Dept(DeptId).

What does the following SQL query give? (Mark all the appropriate options)

```
SELECT Emp.EmpId, Emp.EmpName FROM Emp,Dept
    WHERE Emp.DeptNo = Dept.DeptId
    group by Dept.DeptId having Dept.DeptName = 'ECE'
```

A. It gives the id, name of the employees who works for 'ECE' department
B. It gives the name of the employees whose DeptNo is 2
C. It gives the id,name of the employees whose DeptNo is either 2 or 3
D. The query has error

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following relation.

Emp

EmpId	DeptNo	EmpName	Salary
1	5	Alex	10,000
2	2	Steve	20,000
3	1	Bob	15,000
4	2	Smith	18,000
5	4	Rakul	24,000
6	3	Rhea	17,000

Dept

DeptId	DeptName
1	CSE
2	ECE
3	ECE
4	IT
5	ME

Here, Emp(DeptNo) is the foreign key referring to Dept(DeptId).

The number of tuples returned by the following SQL query is ____

```
SELECT max(Salary) FROM Emp, Dept
    WHERE Emp.DeptNo = Dept.DeptId group by Dept.DeptId;
```

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a $1:n$ relationship and an ER model using it. Which of the following is/are correct for converting this relationship to a relational instance? (Mark all the appropriate options)

A. If participation of both the entities are mandatory then we need only 1 relation
B. If participation of "one-end" is mandatory then we need only 2 relations
C. If participation of "one-end" is optional then we need only 2 relations
D. If participation of both the entities are optional then we need 3 relations

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The following table has two attributes A and B where A is the primary key and B is the foreign key referencing A with on-delete cascade.

A	B
3	2
4	6
1	9
7	4
6	5
11	7
5	6
9	4
8	6

The set of all tuples that must be additionally deleted to preserve the referential integrity when the tuple $(4,6)$ is deleted is _____

A. $(7,4),(9,4),(1,9)$ and $(11,7)$
B. $(6,5),(5,6)$ and $(8,5)$
C. $(7,4),(11,7)$ and $(1,9)$
D. $(7,4),(9,4),(8,6),(4,6)$ and $(5,6)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relation scheme $R = (A, B, C, D, E)$ on which the following functional dependencies hold : $\{AB \rightarrow C, CD \rightarrow E, EA \rightarrow B, ADC \rightarrow B\}$. Which of the following describes all the candidate keys of R?

A. $\{ADB, ADE, ADC\}$
B. $\{AD, ADE, ADC\}$
C. $\{ADB, AE, ADC\}$
D. $\{ADB, ADE, AC\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Mark all the correct statements given below

A. Number of super keys of any relation cannot be more than twice the number of candidate keys
B. The candidate key with the smallest number of attributes is chosen as the primary key
C. A prime attribute must be part of all the candidate keys
D. A prime attribute must be part of some candidate key

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following is a minimal cover of the set of functional dependencies $\{A \rightarrow BC, B \rightarrow CE, A \rightarrow E, AC \rightarrow H, D \rightarrow B\}$?

A. $\{A \rightarrow B, B \rightarrow C, B \rightarrow E, AC \rightarrow H, D \rightarrow B\}$
B. $\{A \rightarrow B, B \rightarrow E, AC \rightarrow H, D \rightarrow B\}$
C. $\{A \rightarrow BH, B \rightarrow CE, D \rightarrow B\}$
D. $\{A \rightarrow B, B \rightarrow C, B \rightarrow E, A \rightarrow H, D \rightarrow B\}$

Your Answer: | Correct Answer: C;D,D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Let $R(A, B, C)$ be a relational schema with the following functional dependencies : $\{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$

The decomposition of R into $(A, B), (B, C)$.

A. Gives a lossless join, and is dependency preserving
B. Gives a lossless join, but is not dependency preserving
C. Does not give a lossless join, but is dependency preserving
D. Does not give a lossless join and is not dependency preserving

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

A table has fields F_1, F_2, F_3, F_4, F_5 with the following functional dependencies: $F_1 \rightarrow F_2, F_2 \rightarrow F_3, F_3 \rightarrow F_4, F_4 \rightarrow F_5, F_1F_2 \rightarrow F_4, F_1F_2 \rightarrow F_5$. The highest normalization form this table is in is _____

A. 1 NF
B. 2 NF
C. 3 NF
D. BCNF

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider relation $R(A, B, C, D, E, F, G, H)$ and the functional dependencies $\{AB \rightarrow C, BD \rightarrow EF, A \rightarrow G, F \rightarrow H\}$. If $2\ NF$ decomposition of R requires minimum $'a'$ relations, and $3\ NF$ decomposition requires minimum $'b'$ relations, the value of $a \times b =$ ____

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following statements concerning database normalization is/are TRUE? (Mark all the appropriate choices)

A. A necessary condition for a relation to be in 1 NF but not in 2 NF is that all its candidate keys must have multiple attributes
B. A necessary condition for a relation to be in 2 NF but not in 3 NF is that some non-prime attribute must be determined by a non-key
C. A necessary condition for a relation to be in 3 NF but not in BCNF is that it should have overlapping candidate keys
D. A necessary condition for a relation to be in 3 NF but not in BCNF is that some prime attribute must be determined by a non-key

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a B-tree in which the search key is 20 bytes long, block size is 1024 bytes, data pointer is 10 bytes and block pointer is 8 Bytes. The maximum order of B-tree will be _____

Your Answer: | Correct Answer: 27 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a $1:1$ relationship and an ER model using it. Which of the following is/are correct for converting this relationship to a relational instance? (Mark all the appropriate options)

A. If participation of both the entities are mandatory then we need 2 relations
B. If participation of both the entities are mandatory then we need only 1 relation

C. If participation of only one of the entity is mandatory then we need 2 relations

D. If participation of both the entities are optional then we need 2 relations

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff